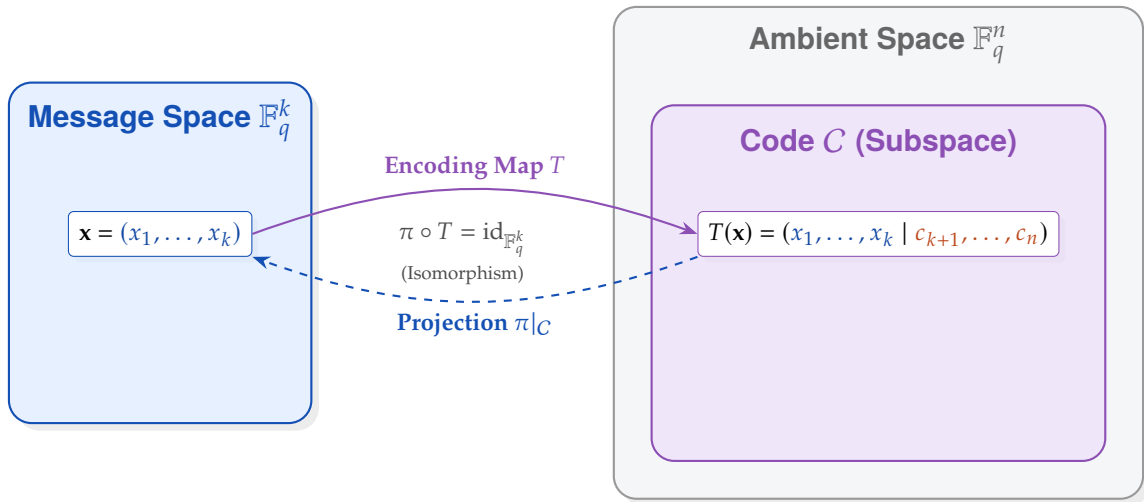




Constructing the Unique Generator Matrix G

Standard Basis in $\mathbb{F}_q^k$

$$\mathbf{e}_1 = (1, 0, \dots, 0)$$

$\vdots$

$$\mathbf{e}_k = (0, 0, \dots, 1)$$

Mapped Codewords in C (Rows of G)

$$T(\mathbf{e}_1) = (1, 0, \dots, 0 \mid a_{1,1}, \dots, a_{1,n-k})$$

$$\vdots \quad G = [I_k \mid A]$$

$$T(\mathbf{e}_k) = (0, 0, \dots, 1 \mid a_{k,1}, \dots, a_{k,n-k})$$